

Grundlagen der Offensiven IT-Sicherheit

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Universität Bonn
WS 2026

Formalities & Setup

1	whoami
2	structure, content & timeline
3	communication channels
4	technical setup

Tip: click any entry to jump to that section

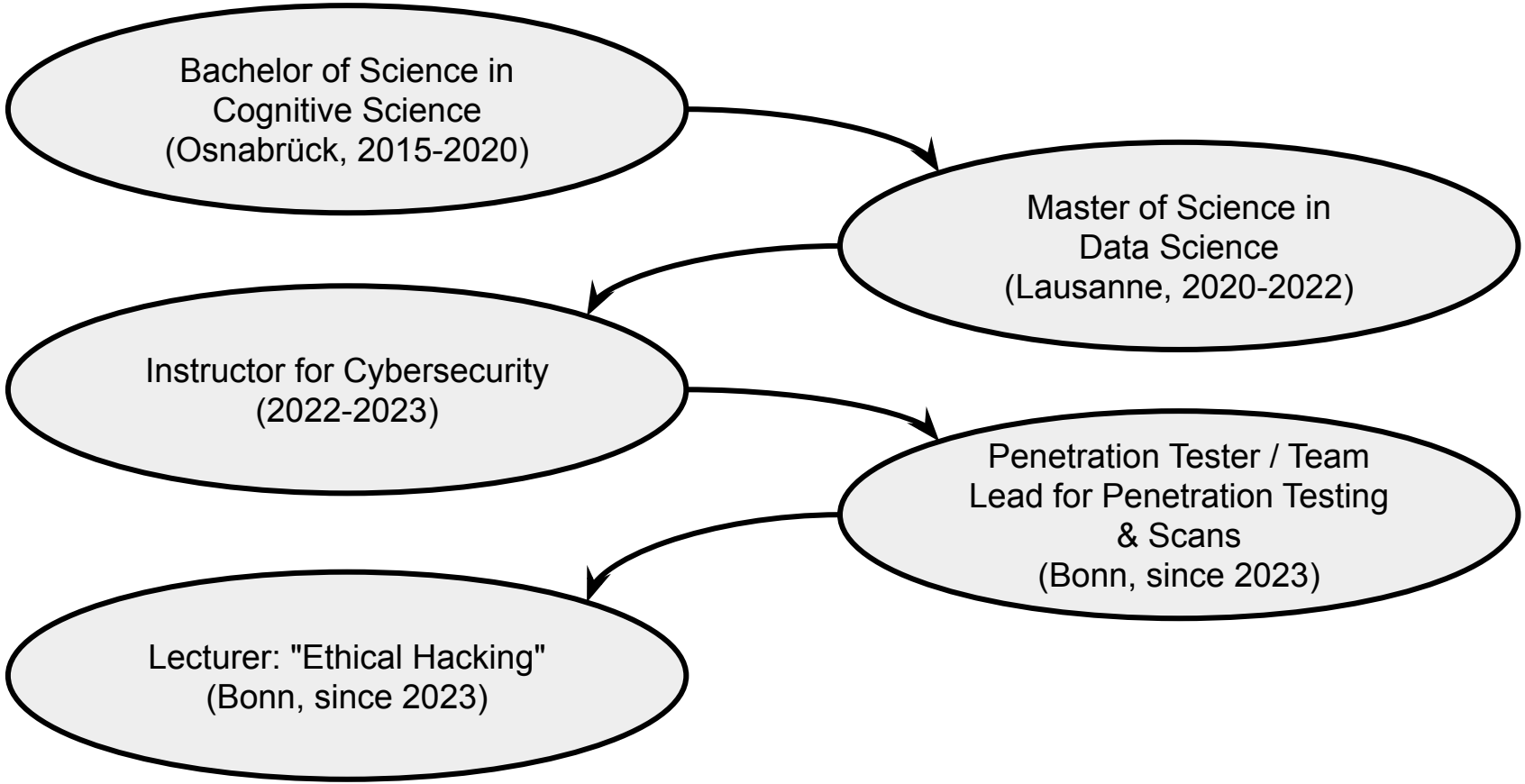
Formalities & Setup

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`whoami`

- really enjoy teaching
  - my goal is to create the best learning experience i can for you
    - i will respect your opinion and your time
    - i'd like that you treat me the same way
  - call me by my first name


Bachelor of Science in
Cognitive Science
(Osnabrück, 2015-2020)
Master of Science in
Data Science
(Lausanne, 2020-2022)
Instructor for Cybersecurity
(2022-2023)
Penetration Tester / Team
Lead for Penetration Testing
& Scans
(Bonn, since 2023)
Lecturer: "Ethical Hacking"
(Bonn, since 2023)

why did you choose this lecture?
what are your expectations?

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General Structure of the Course

- officially, each weekly 3h slot is divided into 1,5h of lecture and 1,5h of exercise
- practically, it will be a bit different
  - 1st part (until christmas holidays): each slot will be 2h-2,5h of lecture, with the rest being used by you to either work on exercise or me to present parts of the previous exercises
  - 2nd part (after new years): slot will be exercises only, with one optional lecture (optional, as in not relevant for the exam)

Building Blocks

lecture

- theory
- me presenting
- fundamentals for hacking

live demos

- me live hacking with commentary and Q&A
- to get a feel about what the thought-process is

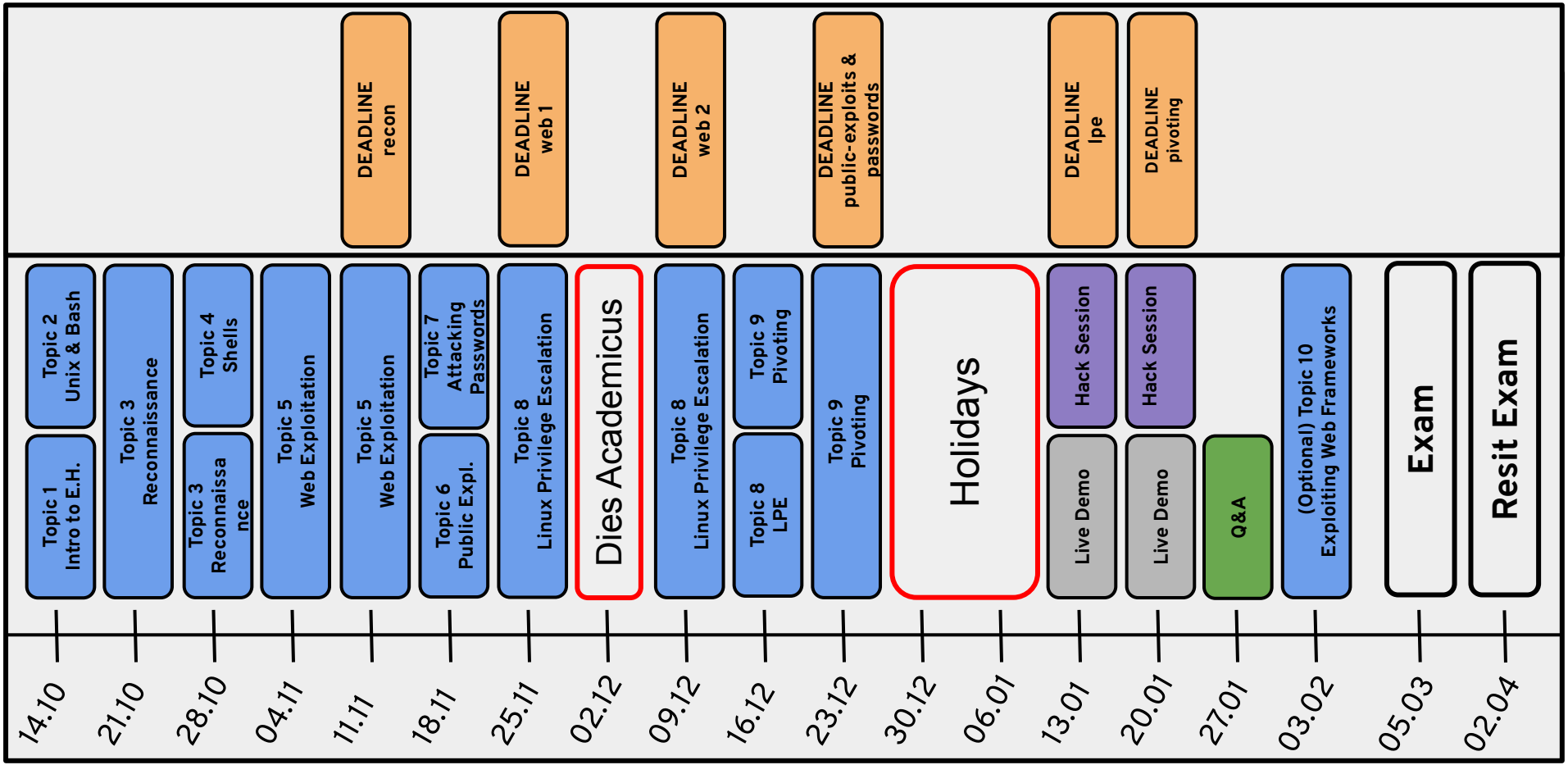
labs

- exercise to apply theory
- designed to help you actually understand how to exploit vulnerabilities
- required for exam admission

hack sessions

- time during lecture time slots to work on the labs
- i will be there for any questions you may have

Timeline


14.10
21.10
28.10
04.11
11.11
18.11
25.11
02.12
09.12
16.12
23.12
30.12
06.01
13.01
20.01
27.01
03.02
05.03
02.04
Topic 1 Intro to E.H.
Topic 2 Unix & Bash
Topic 3 Reconnaissance
Topic 3 Reconnaissance
Topic 4 Shells
Topic 5 Web Exploitation
Topic 5 Web Exploitation
Topic 6 Public Expl.
Topic 7 Attacking Passwords
Topic 8 Linux Privilege Escalation
Dies Academicus
Topic 8 Linux Privilege Escalation
Topic 8 LPE
Topic 9 Pivoting
Topic 9 Pivoting
Holidays
Live Demo
Hack Session
Live Demo
Hack Session
Q&A
(Optional) Topic 10 Exploiting Web Frameworks
Exam
Resit Exam
DEADLINE recon
DEADLINE web 1
DEADLINE web 2
DEADLINE public-exploits & passwords
DEADLINE lpe
DEADLINE pivoting

Exam Admission

- to take the exam for this course, you will have to acquire exam admission
- a total of 8 labs are available to you
  - first lab is an introductory lab for honing you Bash skills
  - last lab belongs to the optional lecture at the end of course, thus is also not exam relevant
  - Information on how the labs are set upsee course website for more concrete information here
- from the remaining six labs, you will have to do five
  - as proof of completion, the deliverable is a write-up
    - regarding what you have to exactly do and submit, more information will follow
    - deadline for each lab is three weeks after its topic's lecture slot
  - the exact dates and rules regarding submissions can be found here

Weekly Tutor Session

- timeslot for the weekly tutor session will be decided until next week's lecture
- please vote and enter you availabilities



Exam

- hand written
- 100% of the course grade
- dates
  - 1st exam: 05.03.2027, 12:00-15:00
  - 2nd exam: 02.04.2027, 12:00-15:00
- combination of
  - multiple choice
  - open questions
  - multi-step exploit riddles
- will test
  - understanding of presented theory
  - exploiting-skills you learned about in the labs and boxes
- exam grades will be standardized based on the class's performance

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Communication Channel - Website



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Winter term 2026/27 at University of Bonn.

New here? Start with Environment Setup, then the Lab Guide.

#	Slides	Cheat sheet	Lab
0	Formalities & Setup	–	–
1	Introduction to Ethical Hacking	–	–
2	Linux and Bash Basics	–	bash
3	Reconnaissance	–	not yet released
4	Shells for the Ethical Hacker	–	–
5	Web Exploitation	–	not yet released
6	Public Exploits	–	not yet released
7	Linux Privilege Escalation	–	not yet released
8	Attacking Passwords	–	not yet released
9	Pivoting	–	not yet released

- https://dcelik.net/teaching/offensive-cybersecurity/ws2026/
- used for
  - sharing instructions, slides, cheatsheets, labs, deadlines & mock exam with you
- disclaimer: new concept and still in development

Communication Channel - eCampus



eCampus | Lernplattform der Universität Bonn

- used for
  - submission of deliverables

Communication Channel - Discord



- https://discord.gg/WkKWDSfvz
- used for
  - announcements from me
  - asking and answering questions (lecture, labs, orga)
  - direct communication with me
  - weekly tutor session with tutor (Raphael Schönefeld)

Communication Channel - Mail



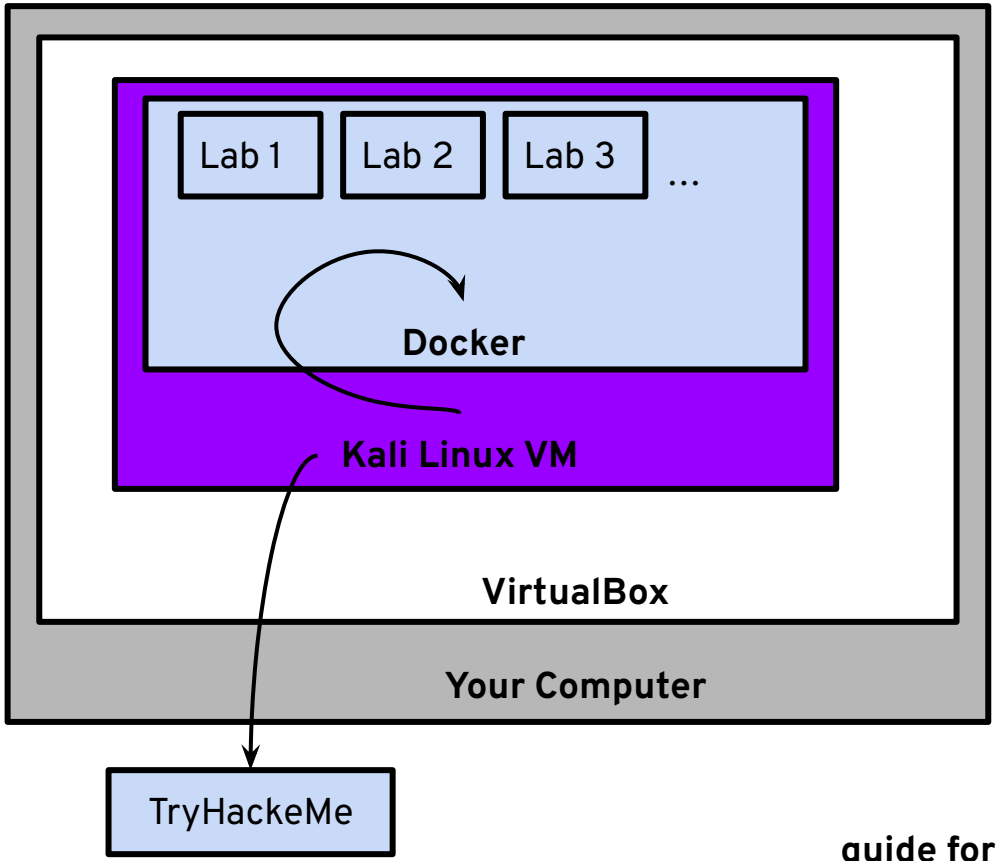
- dcelik@uni-bonn.de
- used for
  - direct communication with me about official/important topics

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Technical Setup



guide for installation and setup was already uploaded to website

Technical Setup

